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Experiment 2: To Containerize Application using Docker

Aim:

To containerize application using Docker

This experiment maps to LO4.

LO4: Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory:

Steps to install docker:

i. Navigate to the official Docker website (docker.com) and download the Docker Desktop installer for your operating system (Windows).
ii. Locate the downloaded installer file on your computer and double-click it to run the setup wizard.
iii. Follow the on-screen prompts. When asked, ensure the option to Use WSL 2 instead of Hyper-V is checked (for optimal performance on Windows).
iv. Wait for the installation to finish, click "Close," and restart your computer if the installer prompts you to do so.
v. After restarting, search for "Docker Desktop" in your system's Start menu and launch the application. Accept the service agreement to start the Docker engine.
vi. Open your terminal or PowerShell and type docker --version and press Enter. If it returns a version number, Docker is successfully installed and ready to use.

Docker:

Docker is an open-source platform that allows you to package an application and all its required dependencies into a single, standardized unit for software development. This ensures that the application runs exactly the same way regardless of where it is deployed—whether on your personal laptop, a testing environment, or a live production server. Instead of dealing with the old problem of "it works on my machine but not yours," Docker isolates the software so it behaves predictably everywhere.

Docker Images: A Docker image is a lightweight, standalone, and immutable (read-only) executable package. In theory, it is a static file that contains the complete set of instructions and components necessary to run a specific application.

An image is built from multiple stacked layers, where each layer represents a specific instruction—such as installing a dependency, setting an environment variable, or copying

application code. It encompasses the application code itself, the runtime environment, system tools, system libraries, and default configuration settings. Because it is read-only, an image cannot be changed once it is built.

Docker Containers: A container is the active, runtime execution of a Docker image. It is a standardized, isolated operating environment where the application actually runs and processes data.

In theory, when an image is executed, the Docker engine adds a thin, writable layer on top of the static image layers. This writable layer allows the active container to modify files, generate logs, and maintain a temporary state while it is running. Containers isolate the running software from the host operating system and from other containers, which ensures the application performs uniformly and predictably regardless of the underlying infrastructure.

Implementation:

1. docker --version

```
C:\Users\Lab203>docker --version
Docker version 29.6.1, build 8900f1d
```

2. docker search ubuntu

```
C:\Users\15L>docker search ubuntu
NAME                              DESCRIPTION                                     STARS     OFFICIAL
ubuntu                            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                      BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                      Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                  ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                    Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus                 Prometheus is a systems and service monitori…   80
ubuntu/jre                        Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet              Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                     A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                   PostgreSQL is an open source object-relation…   45
ubuntu/redis                      Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-runtime             Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps                Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                    Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                  Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                        Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                   Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                      Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                       Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                    Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector    This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager               Ubuntu Rock for Alertmanager. It handles ale…   1
```

3. search ubuntu --filter stars=5

```
C:\Users\15L>docker search ubuntu --filter stars=50
NAME                DESCRIPTION                                     STARS     OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid        Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx        Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9        BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka        Apache Kafka, a distributed event streaming …   62
ubuntu/apache2      Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus   Prometheus is a systems and service monitori…   80
ubuntu/mysql        MySQL open source fast, stable, multi-thread…   74
```

4. docker images

```
C:\Users\Lab203>docker images
                                                                    i Info →  U In Use
IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     ef54e839ef54       25.9kB         9.52kB    U
my-python-app:latest   5bb0461970dd        177MB         43.2MB    U
myubuntu:latest        dbe9a0f59062       1.43GB          390MB    U
nginx:latest           5a88c9c454479       241MB           66MB    U
ubuntu:latest          3131b4cc82a7        161MB         45.3MB    U
```

5. docker search postgres

```
C:\Users\Lab203>docker search postgres
NAME                      DESCRIPTION                                     STARS     OFFICIAL
postgres                  The PostgreSQL object-relational database sy…   14969     [OK]
cimg/postgres                                                             9
circleci/postgres         The PostgreSQL object-relational database sy…   35
kasmweb/postgres          Postgres image maintained by Kasm Technologi…   6
elestio/postgres          Postgres, verified and packaged by Elestio      2
ubuntu/postgres           PostgreSQL is an open source object-relation…   45
cleanstart/postgres       Secure by Design, Built for Speed, Hardened …   0
artifacthub/postgres                                                      0
geokrety/postgres         Postgres with postgis + quantile and amqp ex…   0
corpusops/postgres        https://github.com/corpusops/docker-images/     0
rootpublic/postgres                                                       0
dockette/postgres         My PostgreSQL image with tunning and preinst…   1
vulhub/postgres                                                           2
wayofdev/postgres                                                         0
pgrouting/postgres         Postgres Docker images with PostGIS and dep…   0
openeuler/postgres                                                        0
uselagoon/postgres                                                        0
clarinpl/postgres                                                         0
mosipdev/postgres                                                         0
supabase/postgres         Unmodified Postgres with some useful plugins…   78
brimstone/postgres        postgres image with traefik-cert support        0
trainlineeurope/postgres  Extended version of official Postgres https:…   0
blacklabelops/postgres    Postgres Image for Atlassian Applications       4
fredboat/postgres         PostgreSQL 10.0 used in FredBoat's docker-co…   1
sushantdhiman/postgres    Postgres Image for Sequelize                    2
```

6. docker search --filter stars=100 ubuntu

```
C:\Users\Lab203>docker search --filter stars=100 ubuntu
NAME              DESCRIPTION                                     STARS     OFFICIAL
ubuntu            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx      Nginx, a high-performance reverse proxy & we…   141
ubuntu/apache2    Apache, a secure & extensible open-source HT…   104
ubuntu/bind9      BIND 9 is a very flexible, full-featured DNS…   120
```

7. docker run -it ubuntu /bin/bash

```
C:\Users\Lab203>docker run -it ubuntu /bin/bash
root@8f61b2760bb7:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@8f61b2760bb7:/# exit
exit
```

8. mkdir docker-demo
cd docker-demo

```
C:\Users\Lab203>mkdir docker-demo

C:\Users\Lab203>cd docker demo
The system cannot find the path specified.

C:\Users\Lab203>cd docker-demo
```

9. docker build -t flask-app .

```
C:\Users\Lab203\docker-demo>docker build -t flask-app .
[+] Building 2.4s (10/10) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 212B                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                 0.3s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/5] FROM docker.io/library/python:3.12-slim@sha256:57cd7c3a7a273101a6485ba99423ee568157882804b1124b4dd04266   0.0s
 => => resolve docker.io/library/python:3.12-slim@sha256:57cd7c3a7a273101a6485ba99423ee568157882804b1124b4dd04266   0.0s
 => [internal] load build context                                                                                   0.0s
 => => transferring context: 276B                                                                                   0.0s
 => CACHED [2/5] WORKDIR /app                                                                                       0.0s
 => [3/5] COPY requirements.txt .                                                                                   0.0s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                        1.4s
 => [5/5] COPY . .                                                                                                  0.0s
 => exporting to image                                                                                              0.5s
 => => exporting layers                                                                                             0.3s
 => => exporting manifest sha256:4928296369d6fb986a78476650ded93b3ced5e77e8601f60299e881a99861b1d                   0.0s
 => => exporting config sha256:5025c18b536e9d72b1405e7fc3f2cee7ba957fbf220d527846b28080f439b4ce                     0.0s
 => => exporting attestation manifest sha256:93a09b437c2868ff8683b7a2bea652f5be81d706030adb48c1034beac928cead       0.0s
 => => exporting manifest list sha256:60c2df1030ed44ea46fed60c13532cbdcb0991f20b32c8aa473998d57f4c6df1              0.0s
 => => naming to docker.io/library/flask-app:latest                                                                 0.0s
 => => unpacking to docker.io/library/flask-app:latest                                                              0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/gpu0b6aatfr4jlhblmd9kiuxw
```

10. docker ps

```
C:\Users\Lab203\docker-demo>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

11. docker ps -a

```
C:\Users\Lab203\docker-demo>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS                      PORTS     NAMES
c154cc9048db   flask-app      "python app.py"          48 seconds ago  Exited (0) 48 seconds ago             laughing_grothendieck
f21e0ab9b2f8   flask-app      "python app.py"          5 minutes ago   Exited (0) 5 minutes ago              laughing_zhukovsky
8f61b2760bb7   ubuntu         "/bin/bash"              27 minutes ago  Exited (0) 19 minutes ago             pedantic_edison
e349e1234128   my-python-app  "python app.py"          27 hours ago    Exited (0) 27 hours ago               stoic_swirles
a700eba76fe1   myubuntu       "/bin/bash"              27 hours ago    Exited (127) 27 hours ago             friendly_rhodes
368f82d23c91   myubuntu       "/bin/bash"              27 hours ago    Exited (0) 27 hours ago               nostalgic_snyder
4f79701bdf0a   ubuntu         "/bin/bash"              27 hours ago    Exited (127) 27 hours ago             peaceful_ishizaka
30b568016772   ubuntu         "/bin/bash"              28 hours ago    Exited (255) 35 minutes ago           cool_meitner
50afc5d495c1   ubuntu         "/bin/bash"              28 hours ago    Exited (0) 28 hours ago               cranky_germain
c8d7d78063ff   nginx          "/docker-entrypoint.…"   46 hours ago    Created                               mynginx
db1e61f5826e   ubuntu         "bash"                   46 hours ago    Exited (127) 46 hours ago             friendly_buck
9252b211eddb   ubuntu         "/bin/bash"              46 hours ago    Exited (2) 46 hours ago               hungry_dhawan
1ec237a42c13   ubuntu         "/bin/bash"              46 hours ago    Exited (0) 46 hours ago               admiring_villani
b7d7f64bf20c   ubuntu         "/bin/bash"              46 hours ago    Exited (127) 46 hours ago             suspicious_bhaskara
96d324b8c9b3   ubuntu         "/bin /bash"             46 hours ago    Created                               ecstatic_ride
7d8611fb4d31   ubuntu         "/bin /bash"             46 hours ago    Created                               interesting_dijkstra
a5f07e90484b   ubuntu         "/bin/bash"              46 hours ago    Exited (0) 46 hours ago               sad_hofstadter
f8662871de4c   hello-world    "/hello"                 4 months ago    Exited (0) 4 months ago               hungry_bohr
cbdae8d0d2f5   hello-world    "/hello"                 4 months ago    Exited (0) 4 months ago               awesome_galileo
```

12. docker run -d -p 5000:5000 flask-app

```
C:\Users\Lab203\docker-demo>docker run -d -p 5000:5000 flask-app
aafbbd359fbf7e1d183c2fc8acab5a795be28df3d776802fac85235854584f6a
```

13. type Dockerfile

```
C:\Users\Lab203\docker-demo>
type Dockerfile
FROM python:3.12-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```

14. docker run --rm flask-app

```
C:\Users\Lab203\docker-demo>docker run --rm flask-app

C:\Users\Lab203\docker-demo>docker logs c154cc9048db
```

15. type app.py

```
C:\Users\Lab203\docker-demo>type app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
C:\Users\Lab203\docker-demo>type Dockerfile
FROM python:3.12-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```

16. type requirements.txt

```
C:\Users\Lab203\docker-demo>type requirements.txt
flask
```

17. docker build --no-cache -t flask-app .

```
C:\Users\Lab203\docker-demo>docker build --no-cache -t flask-app .
[+] Building 6.3s (10/10) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 212B                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                 1.2s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/5] FROM docker.io/library/python:3.12-slim@sha256:57cd7c3a7a273101a6485ba99423ee568157882804b1124b4dd04266   0.0s
 => => resolve docker.io/library/python:3.12-slim@sha256:57cd7c3a7a273101a6485ba99423ee568157882804b1124b4dd04266   0.0s
 => [internal] load build context                                                                                   0.0s
 => => transferring context: 477B                                                                                   0.0s
 => CACHED [2/5] WORKDIR /app                                                                                       0.0s
 => [3/5] COPY requirements.txt .                                                                                   0.0s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                        4.3s
 => [5/5] COPY . .                                                                                                  0.0s
 => exporting to image                                                                                              0.7s
 => => exporting layers                                                                                             0.4s
 => => exporting manifest sha256:f8a75af1093a3e99199b2d09f113389a41a391483a16028617b4d450bbe9973e                   0.0s
 => => exporting config sha256:dbc3367f00b00ef53ba2f5b0fe00f505e5ab3e10b0c4bc900eda13b5c455b3d7                     0.0s
 => => exporting attestation manifest sha256:f32030a07a4149402c89fb30a80d4f5206f3cc2c4d835a05653214065c7d96ad       0.0s
 => => exporting manifest list sha256:3379594165db092b6d9604442ee80bfc6eec1de4ac68bc28d95d1e20c62d811a              0.0s
 => => naming to docker.io/library/flask-app:latest                                                                 0.0s
 => => unpacking to docker.io/library/flask-app:latest                                                              0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/yc0774qqa5987wzle39df1nzr
```

18. docker run --rm -p 5000:5000 flask-app

```
C:\Users\Lab203\docker-demo>docker run --rm -p 5000:5000 flask-app
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [22/Jul/2026 03:14:20] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [22/Jul/2026 03:14:20] "GET /favicon.ico HTTP/1.1" 404 -
```



Hello Docker!

Conclusion:

The Docker containerization experiment was carried out successfully by packaging a Flask application into a Docker image using a Dockerfile. The image was built successfully, and a container was created and executed from it. The application's functionality was verified by accessing it through the mapped port and by checking the container status and logs using Docker commands. After successful execution, the container and image were removed, completing the entire container lifecycle. This experiment demonstrated how Docker enables applications and their dependencies to be bundled into a single portable unit, ensuring consistent deployment and execution across different computing environments.